

National Institute of Technology, Kurukshetra
Theory Examination
Question Paper

Month and year of the Examination: **August, 2026**

Programme: **B. Tech**

Semester: II

Subject: Digital System Design

Max. Marks: 50

Course No.: CSIC100

Time: 3:00 Hour

Instructions to candidate:

1. The candidates, before starting to write the solutions, should please check the question paper for any discrepancy, and also ensure also ensure that they have been delivered the question paper of correct course number and subject title.
2. Unless stated otherwise, the symbols have their usual meanings in context with the subject. Assume suitably and state, additional data required, if any.
3. Attempt any FIVE.

Q1	a. Convert decimal number 256 into binary, octal and hexadecimal number system.	2
	b. Subtract $(16)_{10}$ from $(19)_{10}$ using 1's and 2's complement number system.	1
	c. Subtract $(18)_{10}$ from $(16)_{10}$ using 1's and 2's complement number system.	1
	d. Explain following codes	
	i. BCD code	1.5
	ii. Gray code	1.5
	iii. Hamming code	2
Q2	a. Express $f(A, B, C) = A$ in	
	i. Sum of Product form	2
	ii. Product of Sum form	
	b. Minimize following Boolean functions	
	i. $F(w,x,y,z) = \sum 3,7,9,10,13,15 + \sum_d 4,5,6$	2
	ii. $F(w,x,y,z) = \sum 0,2,8,10,16,18,24,26,32,34,40,42,48,50,56,58$ $+ \sum_d (5,7,13,15)$	4
	c. What is rolling of K MAP. Explain.	2
Q3	a. Minimize $F(w,x,y,z) = \sum (1,2,5,6,7,9,10, 11,14)$ using Quine Mc-Cluskey method	5
	b. Explain cyclic and semi cyclic prime implicant table with suitable example.	5
Q4	a. Design and explain 3x8 line decoder. Explain its application in digital circuits.	4
	b. Implement $F(w,x,y,z) = \sum (3,7,9,10,13,15)$ using 8x1 line multiplexer.	3
	c. Design Full adder circuit.	3

Q5	a. Explain following flip flops	
	i. SR flip flop	3
	ii. JK flip flop	3
	iii. Master slave flip flop	4
Q6	Design synchronous modulo 8 counter using T flip flops, Show all steps.	10
Q7	Draw circuit of 4bit ring counter, waveforms of all outputs of flip flops and state diagram. Explain its operation.	10
Q8	Explain fundamental and pulse mode circuits.	10

----- All the best -----

Yes We need it